

SOUMIKA DAS

+91 8617233878 ◇ Kolkata, West Bengal, India

dassoumika11@gmail.com ◇ www.linkedin.com/in/soumikadas ◇ github.com/Soumika-11

SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript, C/C++, Java
- **Web Development:** React, Next.js, HTML, CSS/SCSS, Tailwind, REST APIs, WebSockets
- **Backend:** FastAPI, Microservices Architecture
- **Databases:** PostgreSQL, MySQL/MariaDB
- **AI/ML & Data Science:** Deep Learning, Machine Learning, OpenCV, MediaPipe, YOLO, HuggingFace, Ensemble Learning, Prompt Engineering, Data Preprocessing
- **Hardware & IoT:** Raspberry Pi, Arduino, ESP, Embedded Systems, IoT Prototyping
- **DevOps & Tools:** Git, GitHub, Docker, Docker Compose, Linux

EXPERIENCE

Fullstack Developer Intern Jun 2025 - Sep 2025
Cardiacrhythm Remote

- Developed and maintained backend services with FastAPI & PostgreSQL, improving data processing speed by 30%.
- Enhanced frontend UI/UX in React & TypeScript, boosting user satisfaction by 25% (internal testing feedback).
- Streamlined medical data handling, reducing data retrieval latency by 20%.

Software Development Engineer Intern Jan 2025 - Jun 2025
RevMaxx Onsite

- Built real-time media streaming & AI analysis systems with WebSockets, improving streaming latency by 35%.
- Designed document parsing pipelines that reduced manual screening time by 40%.
- Applied prompt engineering to optimize LLM outputs, achieving a 15% improvement in response accuracy.
- Participated in QA & bug resolution, cutting reported issues by 20% before release.

Research Intern Dec 2024 - Mar 2025
IEM IEDC Hybrid

- Researched and co-authored a conference paper on Brain Tumor Detection using CNN & LSTM, improving model accuracy by 18% over baseline.
- Conducted experiments on Raspberry Pi for AI-enabled edge computing, achieving 10–15% faster inference compared to cloud-only setups.
- Enhanced deep learning models with ensemble methods, boosting diagnostic reliability by 12%.

PROJECTS & PUBLICATIONS

- **Brain Tumor Detector (Conference Paper – AISC):** Developed a deep learning model (CNN, LSTM, Dense, RNN) for MRI-based brain tumor classification. Enhanced performance using data augmentation and presented findings at AISC.
- **Video Proctor:** Built a real-time video proctoring system using OpenCV, MediaPipe & WebSockets. Implemented facial & pose detection for monitoring exam integrity.
- **Resume Parser:** Designed a resume parsing engine to convert PDF/DOCX files into structured JSON for automated recruitment systems.
- **Snomed RAG Medical Coding Scrapper:** Developed a Node.js/TypeScript scraper to fetch and process latest SNOMED medical codes for healthcare applications.

EDUCATION

Bachelor of Technology in Computer Science and Engineering Expected 2027
(IoT, Cybersecurity and Blockchain Technology)
Institute of Engineering and Management, Newtown